

I2C FloorPlan Assignment

Floorplan script:

```
#####
#       ASIC PHYSICAL DESIGN AUTOMATION SCRIPT       #
#####
# Author      : Khaled Gamal
# Project     : I2C
# Block/Top   : i2c_master_top
# Technology  : saed90nm
# EDA Tool    : Synopsys (ICC2)
# Usage       : icc2_shell -f script/floorplan.tcl -output_log
log/floorplan.log
# Prerequisites : Design Library created
# Creation Date : 2026-08-22
# Educational   : ITI_summer_camp_2026_R3
#####

#####
# ----- Variable setup ----- #
#####

set design "i2c"
set std_worst "saed90nm_max.db"
set target_library $std_worst
set corner "worst"

set project_dir "/mnt/hgfs/Shared_files/${design}"
set Lib_path "/home/ICer/Downloads/Lib"
set dlib_path "${project_dir}/pnr/2_design_lib/library"

set prev_stage "dlib"
set current_stage "floorplan"

#####
# ----- dlib setup ----- #
```

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#####

open_block  ${dlib_path}/${design}.dlib:${design}_${prev_stage}.design

# copying the (clean) block from the previous stage (dlib) to just
preserve it and work on the new one which is the present stage (floorplan)
#to make sure if any errors happened through the current stage, it will be
easy to restore the previous stage
copy_block -from_block ${design}.dlib:${design}_${prev_stage}.design
-to_block ${design}_${current_stage}

current_block ${design}_${current_stage}.design
start_gui

#####
# ----- Layers setup ----- #
#####
# Metal layers directions
set_attribute [get_layer {M1 M3 M5 M7 M9}] routing_direction horizontal
set_attribute [get_layer {M2 M4 M6 M8 MRDL}] routing_direction vertical;
# MRDL (Metal Redistribution Layer) is an extra-thick layer on the
absolute top of the die used for power supply

# site def attribute (get unit tile name from the technology file)
set Name_unit [get_site_defs]
set_attribute $Name_unit is_default true

# flipping the unit tiles on the Y-axis to make common Vdd and Vss between
every two site rows
set_attribute $Name_unit symmetry {Y}

# set_attribute [get_layers {M1}] track_offset 0.037

#####
# ----- Initialize Floorplan ----- #
#####

# All parameters related to core and die
initialize_floorplan      -control_type core \
                           -core_utilization 0.6 \
```

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        -shape R \
        -core_offset {10} \
        -flip_first_row true \
        -side_ratio {1 1}

#####
# ----- Placement Pins ----- #
#####
# Auto pins placement (not preferred, manual is better)
place_pins -self -ports [get_ports *]

#####
# ----- Blockage Placement ----- #
#####
#create_placement_blockage -boundary {{30 30} {50 50}} -name B1 -type hard

#create_placement_blockage -boundary {{5 3} {7 5}} -name B2 -type partial
-blockage_percentage 40

#create_placement_blockage -boundary {{7 3} {9 5}} -name B3 -type soft

#####
# ----- TAPCell Placement ----- #
#####
#create_tap_cells -lib_cell [get_lib_cell */SAEDVT14_TAPDS] -pattern
stagger -distance 40

#sizeof_collection [get_cells tap*]
# remove_cell tap*

#create_tap_cells -lib_cell [get_lib_cell */SAEDVT14_TAPDS] -pattern
every_row -distance 10
#get_cells tap*
# remove_cell tap*

#create_tap_cells -lib_cell [get_lib_cell */SAEDVT14_TAPDS] -pattern
every_other_row -distance 10
#size_of[get_cells tap*]
# remove_cell tap*

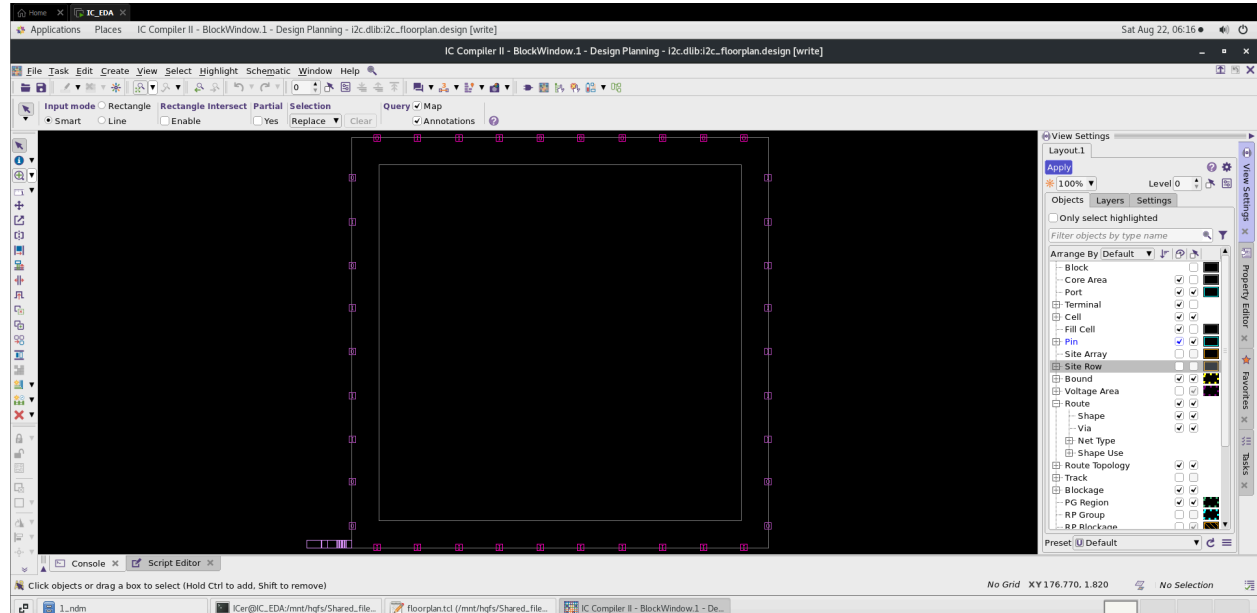
```

```
#####
# ----- Files Handling ----- #
#####
sh rm      -r ../results/reports
sh rm      -r ../results/outputs
sh mkdir -p ../results/reports
sh mkdir -p ../results/outputs

#####
# ----- Reports ----- #
#####
report_gor          > ../results/reports/qor.rpt
report_utilization  > ../results/reports/utilization_default.rpt
get_placement_blockages > ../results/reports/blockage.rpt

#####
# ----- Save Block ----- #
#####
save_block -as ${design}_${current_stage}
${design}.dlib:${design}_${current_stage}.design
```

GUI screenshot:



Reports:

- Blockage report:

Warning: No placement_blockage objects matched '*' (SEL-004)

Error: Nothing matched for collection (SEL-005)

- Qor report:

Report : qor

Design : i2c_master_top

Version: O-2018.06-SP1

Date : Sat Aug 22 06:08:22 2026

Timer Settings:

Delay Calculation Style: auto

Signal Integrity Analysis: disabled

Timing Window Analysis: disabled

Advanced Waveform Propagation: disabled

Variation Type: fixed_derate

Clock Reconvergence Pessimism Removal: disabled

Advanced Receiver Model: disabled

Information: Corner default: no PVT mismatches. (PVT-032)
 Information: The stitching and editing of coupling caps is turned OFF for design 'i2c.dlib:i2c_floorplan.design'. (TIM-125)
 Information: Design Average RC for design i2c_floorplan (NEX-011)
 Information: r = 0.515372 ohm/um, via_r = 0.858630 ohm/cut, c = 0.067760 ff/um, cc = 0.000000 ff/um (X dir) (NEX-017)
 Information: r = 0.562500 ohm/um, via_r = 0.870988 ohm/cut, c = 0.077464 ff/um, cc = 0.000000 ff/um (Y dir) (NEX-017)
 Information: The RC mode used is VR for design 'i2c_master_top'. (NEX-022)
 Information: Update timing completed net estimation for all the timing graph nets (TIM-111)
 Information: Net estimation statistics: timing graph nets = 884, routed nets = 0, across physical hierarchy nets = 0, parasitics cached nets = 0, delay annotated nets = 0, parasitics annotated nets = 0, multi-voltage nets = 0. (TIM-112)

Scenario 'default'
 Timing Path Group 'wb_clk_i'

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Levels of Logic:          9
Critical Path Length:     5.93
Critical Path Slack:      1.55
Critical Path Clk Period: 8.00
Total Negative Slack:     0.00
No. of Violating Paths:   0
Worst Hold Violation:     0.00
Total Hold Violation:     0.00
No. of Hold Violations:   0
-----
  
```

Cell Count

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Hierarchical Cell Count:   4
Hierarchical Port Count:  150
Leaf Cell Count:          807
Buf/Inv Cell Count:       251
Buf Cell Count:           57
Inv Cell Count:           194
CT Buf/Inv Cell Count:    0
Combinational Cell Count: 654
Sequential Cell Count:    153
Macro Count:              0
-----
  
```

Area

Combinational Area:	5676.13
Noncombinational Area:	4827.34
Buf/Inv Area:	1887.44
Total Buffer Area:	670.92
Total Inverter Area:	1216.51
Macro/Black Box Area:	0.00
Net Area:	0
Net XLength:	0.00
Net YLength:	0.00

Cell Area (netlist):	10503.48
Cell Area (netlist and physical only):	10503.48
Net Length:	0.00

Design Rules

Total Number of Nets:	889
Nets with Violations:	0
Max Trans Violations:	0
Max Cap Violations:	0

- Utilization report:

Report : report_utilization

Design : i2c_master_top

Version: O-2018.06-SP1

Date : Sat Aug 22 06:08:41 2026

Utilization Ratio: 0.6132

Utilization options:

- Area calculation based on: site_row of block i2c_floorplan
- Categories of objects excluded: hard_macros macro_keepouts soft_macros io_cells

hard_blockages

Total Area: 17127.9360

Total Capacity Area: 17127.9360

Total Area of cells: 10503.4752

Area of excluded objects:

- hard_macros : 0.0000
- macro_keepouts : 0.0000
- soft_macros : 0.0000
- io_cells : 0.0000
- hard_blockages : 0.0000

Utilization of site-rows with:

- Site 'unit': 0.6132

0.6132